

Customer Loan Approval and Risk Analytics in Banking Sector

1. Project Overview and Objective

This project analyses customer loan approval and financial risk factors using Excel and Power BI. The objective is to clean, transform, and visualize banking data to identify patterns affecting loan approval decisions and customer risk levels.

Objectives

- Perform data cleaning and transformation using Excel.
 - Analyse loan approval and rejection trends.
 - Identify factors affecting loan approval.
 - Build an interactive Power BI dashboard.
 - Generate business insights for risk management.
 - To understand customer risk behavior using financial and credit-related attributes.
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2. Data Source

- Source: Public Banking Dataset (Kaggle)
 - Domain: Banking & Financial Analytics
 - Timeline: 2018–2025
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3. Problem Statement

The main purpose of this project is to analyse customer loan approval patterns and identify financial risk factors affecting lending decisions in banks.

Business Problems Addressed

- To analyse loan approval and rejection trends.
- To identify the impact of annual income, credit score, debt ratio, and employment status on loan approval.
- To evaluate customer risk levels using risk scores.
- To analyse how loan amount and loan duration influence customer repayment ability.
- To identify high-risk customers for better banking decisions.
- To provide insights that help financial institutions reduce loan default risks.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Application Date	Date	Date of loan application
Age	Numeric	Age of customer
Annual Income	Currency/Numeric	Yearly income of customer
Credit Score	Numeric	Customer credit score
Employment Status	Categorical	Employment type/status
Education Level	Categorical	Educational qualification
Loan Amount	Currency/Numeric	Amount requested as loan
Loan Duration	Numeric	Loan repayment duration in months
Marital Status	Categorical	Marital status of customer
Home Ownership Status	Categorical	Type of home ownership
Monthly Debt Payments	Currency/Numeric	Monthly debt payment amount
Debt-to-Income Ratio	Decimal/ Percentage	Debt ratio compared to income
Loan Purpose	Categorical	Reason for taking loan
Payment History	Numeric	Previous payment history
Savings Account Balance	Currency/Numeric	Savings balance maintained by customer
Monthly Income	Currency/Numeric	Monthly salary/income
Net Worth	Currency/Numeric	Customer net worth
Interest Rate	Percentage	Loan interest rate
Loan Approval	Binary (0/1)	Loan approved or rejected
Risk Score	Numeric	Customer financial risk score

5. Tools & Technologies

Excel

- Data cleaning
- Duplicate removal
- Handling missing values
- Pivot Tables
- Data transformation

Power BI

- DAX calculations
- Dashboard creation
- Interactive visualizations
- KPI cards
- Slicers and filters

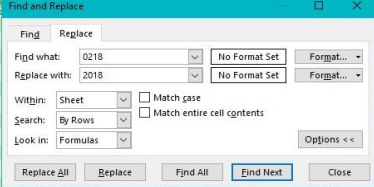
6. Data Pre-Processing (Excel / Power Query)

Tasks Performed

- Removed duplicate records.
- Handled missing values using suitable methods.
- Corrected inconsistent data formats.
- Converted columns into proper data types.
- Standardized date, percentage, and currency formats.
- Created calculated columns
- Used Pivot Tables for analysis

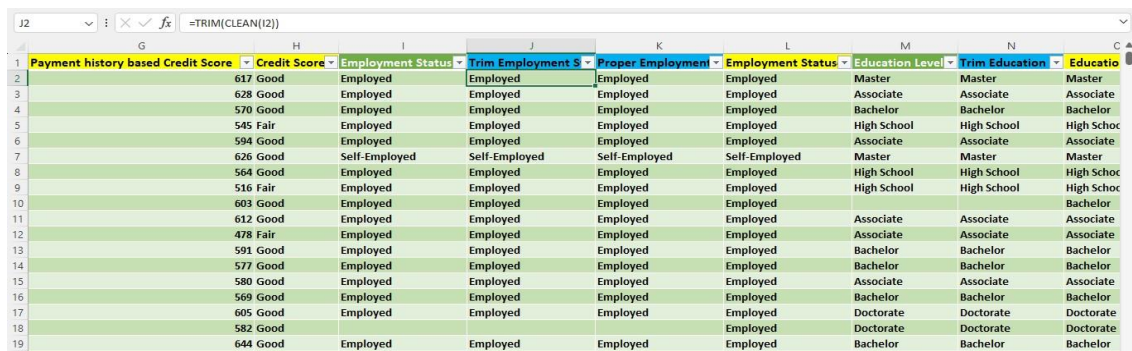
CORRECTED INCONSISTENT DATA FORMATS

FIND & REPLACE



ApplicationDate	Age	AnnualIncome	Credit	Duration	MaritalStatus	HomeOwnershipStatus	MonthlyDebtPayments	Debt
0218-01-09	34	\$1,16,841		60	Divorced	Own		638
0218-03-12	44	\$19,924		48	Married	Mortgage		495
2018-40-04	36	\$1,52,650		48	Married	Other		148
2018-042-0	39	\$29,900		72	Single	Mortgage		589
2018-053-0	43	\$24,611		24	Married	Other		489
2018-06-92	72	\$1,15,677		24	Married	Own		364
2018-0-704	37	\$68,157		36	Married	Mortgage		394
201-807-16	29	\$55,172		36	Married	Mortgage		317
2018-0-725	28	\$1,37,635		48	Divorced	Own		285
2018-0-731	53	\$85,512		48	Single	Own		338
2018-09-28	57				Single	Mortgage		410
2019-020-7	40	\$61,340	617 mEmployed	24	Married	Own		174
201-902-19	43	\$44,674	574	24	Single	Rent		538
2019-0-407	40	\$97,910	649 Employde	48	Married	Other		371
2019-04-09	42	\$84,242	603 Employed	84	Single	Rent		242
2019-041-7	18	\$33,339	578 Epmloyed	72	Single	Mortgage		553
201-905-06	39	\$56,490	609 EMPLOYED	60	Married	Rent		117
2019-60-12	40	\$42,120	547	72	Single	Mortgage		360
20190-6-23	52	\$41,332	565 Employde	24	Married	Rent		248

TRIM



Payment history based Credit Score	Credit Score	Employment Status	Trim Employment S	Proper Employment	Employment Status	Education Level	Trim Education	Educatio
617 Good	Employed	Employed	Employed	Employed	Employed	Master	Master	Master
628 Good	Employed	Employed	Employed	Employed	Employed	Associate	Associate	Associate
570 Good	Employed	Employed	Employed	Employed	Employed	Bachelor	Bachelor	Bachelor
545 Fair	Employed	Employed	Employed	Employed	Employed	High School	High School	High Schoc
594 Good	Employed	Employed	Employed	Employed	Employed	Associate	Associate	Associate
626 Good	Self-Employed	Self-Employed	Self-Employed	Self-Employed	Self-Employed	Master	Master	Master
564 Good	Employed	Employed	Employed	Employed	Employed	High School	High School	High Schoc
516 Fair	Employed	Employed	Employed	Employed	Employed	High School	High School	High Schoc
603 Good	Employed	Employed	Employed	Employed	Employed			Bachelor
612 Good	Employed	Employed	Employed	Employed	Employed	Associate	Associate	Associate
478 Fair	Employed	Employed	Employed	Employed	Employed	Associate	Associate	Associate
591 Good	Employed	Employed	Employed	Employed	Employed	Bachelor	Bachelor	Bachelor
577 Good	Employed	Employed	Employed	Employed	Employed	Bachelor	Bachelor	Bachelor
580 Good	Employed	Employed	Employed	Employed	Employed	Associate	Associate	Associate
569 Good	Employed	Employed	Employed	Employed	Employed	Bachelor	Bachelor	Bachelor
605 Good	Employed	Employed	Employed	Employed	Employed	Doctorate	Doctorate	Doctorate
582 Good	Employed	Employed	Employed	Employed	Employed	Doctorate	Doctorate	Doctorate
644 Good	Employed	Employed	Employed	Employed	Employed	Bachelor	Bachelor	Bachelor

PROPER

=PROPER(J2)									
Trim Employment S	Proper Employment	Employment Status	Education Level	Trim Education	Education Leve	Loan Amount	Loan Amount	Loan to Incon	Borrower Profile
Employed	Employed	Employed	Master	Master	Master	\$13,152	\$13,152	33%	Conservative Borrow
Employed	Employed	Employed	Associate	Associate	Associate	\$26,045	\$26,045	66%	Moderate Borrower
Employed	Employed	Employed	Bachelor	Bachelor	Bachelor	\$17,627	\$17,627	43%	Conservative Borrow
Employed	Employed	Employed	High School	High School	High School	\$37,898	\$37,898	55%	Moderate Borrower
Employed	Employed	Employed	Associate	Associate	Associate	\$9,184	\$9,184	9%	Conservative Borrow
Self-Employed	Self-Employed	Self-Employed	Master	Master	Master	\$15,433	\$15,433	9%	Conservative Borrow
Employed	Employed	Employed	High School	High School	High School	\$12,741	\$12,741	25%	Conservative Borrow
Employed	Employed	Employed	High School	High School	High School	\$19,634	\$19,634	20%	Conservative Borrow
Employed	Employed	Employed	Associate	Associate	Associate	\$55,353	\$55,353	47%	Conservative Borrow
Employed	Employed	Employed	Associate	Associate	Associate	\$25,443	\$25,443	63%	Moderate Borrower
Employed	Employed	Employed	Associate	Associate	Associate	\$48,716	\$48,716	66%	Moderate Borrower
Employed	Employed	Employed	Bachelor	Bachelor	Bachelor	\$30,088	\$30,088	201%	Aggressive Borrower
Employed	Employed	Employed	Bachelor	Bachelor	Bachelor	\$16,154	\$16,154	22%	Conservative Borrow
Employed	Employed	Employed	Associate	Associate	Associate	\$20,439	\$20,439	20%	Conservative Borrow
Employed	Employed	Employed	Bachelor	Bachelor	Bachelor	\$27,197	\$27,197	57%	Moderate Borrower
Employed	Employed	Employed	Doctorate	Doctorate	Doctorate	\$12,652	\$12,652	22%	Conservative Borrow
		Employed	Doctorate	Doctorate	Doctorate		\$25,184	50%	Moderate Borrower
Employed	Employed	Employed	Bachelor	Bachelor	Bachelor		\$24,648	17%	Conservative Borrow
Employed	Employed	Employed	High School	High School	High School	\$14,655	\$14,655	91%	Moderate Borrower

HANDLED MISSING VALUES USING SUITABLE METHODS

Annual Income

=IF(D2="",AC2*12,D2)									
Application Date	Age	AGE GROUP	Annual Income	Annual Income	Credit Score	Payment history based Credit Score	Credit Score	Employment Status	Trim Employment S
01-01-2018	45	Senior	\$39,948	\$39,948	617	617	Good	Employed	Employed
02-01-2018	38	Mid	\$39,709	\$39,709	628	628	Good	Employed	Employed
03-01-2018	47	Senior	\$40,724	\$40,724	570	570	Good	Employed	Employed
04-01-2018	58	Senior	\$69,084	\$69,084	545	545	Fair	Employed	Employed
05-01-2018	37	Mid	\$1,03,264	\$1,03,264	594	594	Good	Employed	Employed
06-01-2018	37	Mid	\$1,78,310	\$1,78,310	626	626	Good	Self-Employed	Self-Employed
07-01-2018	58	Senior	\$51,250	\$51,250	564	564	Good	Employed	Employed
08-01-2018	49	Senior	\$97,345	\$97,345	516	516	Fair	Employed	Employed
09-01-2018	34	Mid	\$1,16,841	\$1,16,841	603	603	Good	Employed	Employed
10-01-2018	46	Senior	\$40,615	\$40,615	612	612	Good	Employed	Employed
11-01-2018	34	Mid		\$73,646	478	478	Fair	Employed	Employed
12-01-2018	34	Mid	\$15,000	\$15,000	591	591	Good	Employed	Employed
13-01-2018	42	Senior	\$74,453	\$74,453		577	Good	Employed	Employed
14-01-2018	18	Young		\$1,00,508	580	580	Good	Employed	Employed

Credit Score

=IF(F2="",AVERAGEIF(\$AA\$2:\$AA\$2648,AA2,\$F\$2:\$F\$2648),F2)									
Application Date	Age	AGE GROUP	Annual Income	Annual Income	Credit Score	Payment history based Credit Score	Credit Score	Employment Status	Trim Employment S
01-01-2018	45	Senior	\$39,948	\$39,948	617	617	Good	Employed	Employed
02-01-2018	38	Mid	\$39,709	\$39,709	628	628	Good	Employed	Employed
03-01-2018	47	Senior	\$40,724	\$40,724	570	570	Good	Employed	Employed
04-01-2018	58	Senior	\$69,084	\$69,084	545	545	Fair	Employed	Employed
05-01-2018	37	Mid	\$1,03,264	\$1,03,264	594	594	Good	Employed	Employed
06-01-2018	37	Mid	\$1,78,310	\$1,78,310	626	626	Good	Self-Employed	Self-Employed
07-01-2018	58	Senior	\$51,250	\$51,250	564	564	Good	Employed	Employed
08-01-2018	49	Senior	\$97,345	\$97,345	516	516	Fair	Employed	Employed
09-01-2018	34	Mid	\$1,16,841	\$1,16,841	603	603	Good	Employed	Employed
10-01-2018	46	Senior	\$40,615	\$40,615	612	612	Good	Employed	Employed
11-01-2018	34	Mid		\$73,646	478	478	Fair	Employed	Employed
12-01-2018	34	Mid	\$15,000	\$15,000	591	591	Good	Employed	Employed
13-01-2018	42	Senior	\$74,453	\$74,453		577	Good	Employed	Employed
14-01-2018	18	Young		\$1,00,508	580	580	Good	Employed	Employed

Y	Z	AA	AB	A
	Row Labels	Count of Employment Status2		
		170		
	Employed	2111		
	Self-Employed	189		
	Unemployed	177		
	Grand Total	2647		
	Row Labels	Count of Education Level2		
		154		
	Associate	483		
	Bachelor	799		
	Doctorate	118		
	High School	728		
	Master (blank)	365		
	Grand Total	2647		

PivotTable Fields

Choose fields to add to report:

Search

☒ **Education Level2**
More Tables...

Drag fields between areas below:

Filters	Columns
Rows Education Level2	Σ Values Count of Education Lev...

L	M	N	O	P	Q	R	S	T	U
Employment Status	Education Level	Trim Education	Education Level	Loan Amount	Loan Amount	Loan to Income	Borrower Profile	Loan Duration	Loan Duration
2	Employed	Master	Master	\$13,152	\$13,152	33%	Conservative Borrower	48	48
3	Employed	Associate	Associate	\$26,045	\$26,045	66%	Moderate Borrower	48	48
4	Employed	Bachelor	Bachelor	\$17,627	\$17,627	43%	Conservative Borrower	36	36
5	Employed	High School	High School	\$37,898	\$37,898	55%	Moderate Borrower	96	96
6	Employed	Associate	Associate	\$9,184	\$9,184	9%	Conservative Borrower	36	36
7	Self-Employed	Master	Master	\$15,433	\$15,433	9%	Conservative Borrower		51
8	Employed	High School	High School	\$12,741	\$12,741	25%	Conservative Borrower	48	48
9	Employed	High School	High School	\$19,634	\$19,634	20%	Conservative Borrower	12	12
10	Employed		Bachelor	\$55,353	\$55,353	47%	Conservative Borrower	60	60
11	Employed	Associate	Associate	\$25,443	\$25,443	63%	Moderate Borrower	12	12
12	Employed	Associate	Associate	\$48,716	\$48,716	66%	Moderate Borrower		54
13	Employed	Bachelor	Bachelor	\$30,088	\$30,088	201%	Aggressive Borrower	24	24

	L	M	N	O	P	Q	R	S	T	U
1	Employment Status	Education Level	Trim Education	Education Level	Loan Amount	Loan Amount	Loan to Incon	Borrower Profile	Loan Duration	Loan Duration
2	Employed	Master	Master	Master	\$13,152	\$13,152	33%	Conservative Borrower	48	48
3	Employed	Associate	Associate	Associate	\$26,045	\$26,045	66%	Moderate Borrower	48	48
4	Employed	Bachelor	Bachelor	Bachelor	\$17,627	\$17,627	43%	Conservative Borrower	36	36
5	Employed	High School	High School	High School	\$37,898	\$37,898	55%	Moderate Borrower	96	96
6	Employed	Associate	Associate	Associate	\$9,184	\$9,184	9%	Conservative Borrower	36	36
7	Self-Employed	Master	Master	Master	\$15,433	\$15,433	9%	Conservative Borrower	53	53
8	Employed	High School	High School	High School	\$12,741	\$12,741	25%	Conservative Borrower	48	48
9	Employed	High School	High School	High School	\$19,634	\$19,634	20%	Conservative Borrower	12	12
10	Employed			Bachelor	\$55,353	\$55,353	47%	Conservative Borrower	60	60
11	Employed	Associate	Associate	Associate	\$25,443	\$25,443	63%	Moderate Borrower	12	12
12	Employed	Associate	Associate	Associate	\$48,716	\$48,716	66%	Moderate Borrower	54	54
13	Employed	Bachelor	Bachelor	Bachelor	\$30,088	\$30,088	201%	Aggressive Borrower	24	24
14	Employed	Bachelor	Bachelor	Bachelor	\$16,154	\$16,154	22%	Conservative Borrower	60	60

	M	N	O	P	Q	R	S	T	U	V
1	Education Level	Trim Education	Education Level	Loan Amount	Loan Amount	Loan to Income	Borrower Profile	Loan Duration	Loan Duration	Marital Status
2	Master	Master	Master	\$13,152	\$13,152	33%	Conservative Borrower	48	48	Married
3	Associate	Associate	Associate	\$26,045	\$26,045	66%	Moderate Borrower	48	48	Single
4	Bachelor	Bachelor	Bachelor	\$17,627	\$17,627	43%	Conservative Borrower	36	36	Married
5	High School	High School	High School	\$37,898	\$37,898	55%	Moderate Borrower	96	96	Single
6	Associate	Associate	Associate	\$9,184	\$9,184	9%	Conservative Borrower	36	36	Married
7	Master	Master	Master	\$15,433	\$15,433	9%	Conservative Borrower		53	Married
8	High School	High School	High School	\$12,741	\$12,741	25%	Conservative Borrower	48	48	Married
9	High School	High School	High School	\$19,634	\$19,634	20%	Conservative Borrower	12	12	Divorced
10			Bachelor	\$55,353	\$55,353	47%	Conservative Borrower	60	60	Divorced
11	Associate	Associate	Associate	\$25,443	\$25,443	63%	Moderate Borrower	12	12	Married
12	Associate	Associate	Associate	\$48,716	\$48,716	66%	Moderate Borrower		54	Single
13	Bachelor	Bachelor	Bachelor	\$30,088	\$30,088	201%	Aggressive Borrower	24	24	Married
14	Bachelor	Bachelor	Bachelor	\$16,154	\$16,154	22%	Conservative Borrower	60	60	Married
15	Associate	Associate	Associate	\$20,439	\$20,439	20%	Conservative Borrower		53	Married

Feature Engineering

In this project, several new derived features were created to improve analysis and model understanding:

- **Age Group** – Age values were categorized into groups such as Young, Mid and Senior to simplify demographic analysis.

Application Date	Age	AGE GROUP	Annual Income	Annual Income	Credit Score	Payment history based Credit Score	Credit Score	Employment Status	Trim Employment S
01-01-2018	45	Senior	\$39,948	\$39,948	617	617 Good	617 Good	Employed	Employed
02-01-2018	38	Mid	\$39,709	\$39,709	628	628 Good	628 Good	Employed	Employed
03-01-2018	47	Senior	\$40,724	\$40,724	570	570 Good	570 Good	Employed	Employed
04-01-2018	58	Senior	\$69,084	\$69,084	545	545 Fair	545 Fair	Employed	Employed
05-01-2018	37	Mid	\$1,03,264	\$1,03,264	594	594 Good	594 Good	Employed	Employed
06-01-2018	37	Mid	\$1,78,310	\$1,78,310	626	626 Good	626 Good	Self-Employed	Self-Employed
07-01-2018	58	Senior	\$51,250	\$51,250	564	564 Good	564 Good	Employed	Employed
08-01-2018	49	Senior	\$97,345	\$97,345	516	516 Fair	516 Fair	Employed	Employed
09-01-2018	34	Mid	\$1,16,841	\$1,16,841	603	603 Good	603 Good	Employed	Employed
10-01-2018	46	Senior	\$40,615	\$40,615	612	612 Good	612 Good	Employed	Employed
11-01-2018	34	Mid	\$73,646	\$73,646	478	478 Fair	478 Fair	Employed	Employed

- **Credit Score Band** – Credit scores were grouped into categories like Excellent, Fair, Good and Poor for risk segmentation.

Application Date	Age	AGE GROUP	Annual Income	Annual Income	Credit Score	Payment history based Credit Score	Credit Score	Employment Status	Trim Employment S
01-01-2018	45	Senior	\$39,948	\$39,948	617	617 Good	617 Good	Employed	Employed
02-01-2018	38	Mid	\$39,709	\$39,709	628	628 Good	628 Good	Employed	Employed
03-01-2018	47	Senior	\$40,724	\$40,724	570	570 Good	570 Good	Employed	Employed
04-01-2018	58	Senior	\$69,084	\$69,084	545	545 Fair	545 Fair	Employed	Employed
05-01-2018	37	Mid	\$1,03,264	\$1,03,264	594	594 Good	594 Good	Employed	Employed
06-01-2018	37	Mid	\$1,78,310	\$1,78,310	626	626 Good	626 Good	Self-Employed	Self-Employed
07-01-2018	58	Senior	\$51,250	\$51,250	564	564 Good	564 Good	Employed	Employed
08-01-2018	49	Senior	\$97,345	\$97,345	516	516 Fair	516 Fair	Employed	Employed
09-01-2018	34	Mid	\$1,16,841	\$1,16,841	603	603 Good	603 Good	Employed	Employed

- **Loan-to-Income Ratio** – A new feature created by comparing loan amount with income to understand repayment capability.

Education Level	Trim Education	Education Level	Loan Amount	Loan Amount	Loan to Incon	Borrower Profile	Loan Duration	Loan Duration	Marital Status	Hom
Master	Master	Master	\$13,152	\$13,152	33%	Conservative Borrower	48	48	Married	Own
Associate	Associate	Associate	\$26,045	\$26,045	66%	Moderate Borrower	48	48	Single	Mort
Bachelor	Bachelor	Bachelor	\$17,627	\$17,627	43%	Conservative Borrower	36	36	Married	Rent
High School	High School	High School	\$37,898	\$37,898	55%	Moderate Borrower	96	96	Single	Mort
Associate	Associate	Associate	\$9,184	\$9,184	9%	Conservative Borrower	36	36	Married	Mort
Master	Master	Master	\$15,433	\$15,433	9%	Conservative Borrower	53	53	Married	Rent
High School	High School	High School	\$12,741	\$12,741	25%	Conservative Borrower	48	48	Married	Rent
High School	High School	High School	\$19,634	\$19,634	20%	Conservative Borrower	12	12	Divorced	Mort
		Bachelor	\$55,353	\$55,353	47%	Conservative Borrower	60	60	Divorced	Own
Associate	Associate	Associate	\$25,443	\$25,443	63%	Moderate Borrower	12	12	Married	Othe
Associate	Associate	Associate	\$48,716	\$48,716	66%	Moderate Borrower	54	54	Single	Own
Bachelor	Bachelor	Bachelor	\$30,088	\$30,088	201%	Aggressive Borrower	24	24	Married	Mort

- **Borrower Profile** – Combined attributes like income, employment, and credit behaviour to form a customer profile category.
- **Monthly Income Category** – Income was segmented into different ranges for better financial classification.

- **Risk Score Label** – Risk scores were converted into labels such as Low Risk, Medium Risk, and High Risk to support decision-making.

AI2 $\text{=IF(AH2<40,"Low Risk", IF(AH2<55,"Medium Risk", IF(AH2<70,"High Risk","Very High Risk"))}$

	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN
1	Monthly Income	Monthly Income	Net Worth	Interest Rate	Loan Approved	Risk Score	RISK SCORE LABEL					
2	\$3,329 Low Income		\$1,26,928	23%	0	49	Medium Risk					
3	\$3,309 Low Income		\$43,609	20%	0	52	Medium Risk					
4	\$3,394 Low Income		\$5,205	21%	0	52	Medium Risk					
5	\$5,757 Medium Income		\$99,452	30%	0	54	Medium Risk					
6	\$8,605 Medium Income		\$2,27,019	18%	1	36	Low Risk					
7	\$14,859 High Income		\$27,071	22%	1	44	Medium Risk					
8	\$4,271 Low Income		\$21,730	21%	0	50	Medium Risk					
9	\$8,112 Medium Income		\$38,621	21%	1	42	Medium Risk					
10	\$9,737 Medium Income		\$7,711	29%	0	61	High Risk					
11	\$3,385 Low Income		\$1,16,812	20%	0	53	Medium Risk					
12	\$6,137 Medium Income		\$1,28,651	33%	0	47	Medium Risk					
13	\$1,250 Low Income		\$29,709	18%	0	56	High Risk					

Row Labels Count of Employment Status2

Employed	170
Self-Employed	2111
Unemployed	189
Grand Total	2647

Row Labels Count of Education Level2

Associate	154
Bachelor	483
	799

7. DAX (Power BI)

Calculated Measures

- **Total Loan Applications**

1 Total Applications = COUNTROWS('CLEANED DATASET')

	Savings Account Balance	Monthly Income Category	Net Worth	Interest Rate	Loan Approved	Risk Score	Risk Score Label	Loan Status
20	\$886	Low Income	\$5,205	21%	0	52	Medium Risk	Rejected
18	\$1,346	Low Income	\$2,705	16%	0	48	Medium Risk	Rejected
26	\$2,926	Low Income	\$3,13,319	15%	0	50	Medium Risk	Rejected
22	\$4,029	Low Income	\$71,090	20%	0	50	Medium Risk	Rejected
11	\$2,267	Low Income	\$34,774	18%	0	49	Medium Risk	Rejected
25	\$3,815	Low Income	\$2,253	27%	0	52	Medium Risk	Rejected
19	\$7,167	Low Income	\$78,660	22%	0	49	Medium Risk	Rejected
21	\$10,050	Low Income	\$9,540	26%	0	51	Medium Risk	Rejected
29	\$9,920	Low Income	\$4,623	24%	0	50	Medium Risk	Rejected
24	\$13,605	Low Income	\$47,177	23%	0	52	Medium Risk	Rejected
31	\$673	Low Income	\$25,415	23%	0	50	Medium Risk	Rejected
30	\$2,817	Low Income	\$69,075	24%	0	50	Medium Risk	Rejected
26	\$3,099	Low Income	\$2,880	26%	0	54	Medium Risk	Rejected
29	\$11,187	Low Income	\$2,007	34%	0	54	Medium Risk	Rejected
23	\$12,430	Low Income	\$23,478	19%	0	52	Medium Risk	Rejected
26	\$14,928	Low Income	\$48,358	20%	0	52	Medium Risk	Rejected
31	\$2,250	Low Income	\$3,965	23%	0	47	Medium Risk	Rejected
22	\$8,040	Low Income	\$2,610	18%	0	51	Medium Risk	Rejected

Data

Search

- Σ Loan Approved
- Σ Loan Approved Count
- Σ Loan Duration
- Loan Purpose
- Loan Rejected Count
- Loan Status
- Σ Loan to Income Ratio
- Marital Status
- Σ Monthly Debt Payments
- Σ Monthly Income
- Monthly Income Category
- Σ Net Worth
- Σ Payment History
- Σ Risk Score
- Risk Score Label

- **Approved Loans**

1 Loan Approved Count = CALCULATE(COUNTROWS('CLEANED DATASET'),'CLEANED DATASET'[Loan Approved])

	Application Date	Age	Age Group	Annual Income	Credit Score	Credit Score Band	Employment Status	Education Level	Loan Amount	Loan Duration
	03-01-2018	47	Senior	\$40,724	570	Good	Employed	Bachelor	\$17,627	
	29-01-2018	32	Mid	\$35,498	622	Good	Employed	Bachelor	\$11,136	
	09-03-2018	52	Senior	\$32,418	647	Good	Employed	Bachelor	\$32,208	
	10-03-2018	44	Senior	\$26,879	649	Good	Employed	Bachelor	\$20,313	
	17-03-2018	49	Senior	\$40,829	617	Good	Employed	Bachelor	\$25,478	
	01-04-2018	41	Senior	\$51,707	585	Good	Employed	Bachelor	\$20,069	
	08-04-2018	43	Senior	\$30,948	601	Good	Employed	Bachelor	\$25,291	
	26-04-2018	43	Senior	\$49,421	609	Good	Employed	Bachelor	\$17,161	
	28-04-2018	25	Mid	\$41,402	572	Good	Employed	Bachelor	\$25,900	
	05-05-2018	47	Senior	\$50,345	602	Good	Employed	Bachelor	\$18,805	
	01-06-2018	44	Senior	\$47,430	581	Good	Employed	Bachelor	\$32,684	
	22-06-2018	39	Mid	\$48,707	582	Good	Employed	Bachelor	\$9,741	
	05-09-2018	32	Mid	\$34,817	606	Good	Employed	Bachelor	\$36,726	
	21-09-2018	27	Mid	\$29,349	577	Good	Employed	Bachelor	\$31,897	
	30-09-2018	53	Senior	\$45,788	635	Good	Employed	Bachelor	\$31,589	
	28-10-2018	30	Mid	\$26,960	603	Good	Employed	Bachelor	\$12,517	
	12-11-2018	40	Senior	\$36,971	611	Good	Employed	Bachelor	\$29,198	

Data

Search

- Borrower Profile
- Credit Score Band
- Σ Debt To Income Ratio
- Employment Status
- High Risk Customer
- Home Ownership Status
- Σ Interest Rate
- Σ Loan Amount
- Σ Loan Approved
- Σ Loan Approved Count
- Σ Loan Duration
- Loan Purpose
- Loan Rejected Count
- Loan Status

• Rejected Loans

1 Loan Rejected Count = CALCULATE(COUNTROWS('CLEANED DATASET'),'CLEANED DATASET'[Loan Approved]=0)

Application Date	Age	Age Group	Annual Income	Credit Score	Credit Score Band	Employment Status	Education Level	Loan Amount	Loan Duration
03-01-2018	47	Senior	\$40,724	570	Good	Employed	Bachelor	\$17,627	
29-01-2018	32	Mid	\$35,498	622	Good	Employed	Bachelor	\$11,136	
09-03-2018	52	Senior	\$32,418	647	Good	Employed	Bachelor	\$32,208	
10-03-2018	44	Senior	\$26,879	649	Good	Employed	Bachelor	\$20,313	
17-03-2018	49	Senior	\$40,829	617	Good	Employed	Bachelor	\$25,478	
01-04-2018	41	Senior	\$51,707	585	Good	Employed	Bachelor	\$20,069	
08-04-2018	43	Senior	\$30,948	601	Good	Employed	Bachelor	\$25,291	
26-04-2018	43	Senior	\$49,421	609	Good	Employed	Bachelor	\$17,161	
28-04-2018	25	Mid	\$41,402	572	Good	Employed	Bachelor	\$25,900	
05-05-2018	47	Senior	\$50,345	602	Good	Employed	Bachelor	\$18,805	
01-06-2018	44	Senior	\$47,430	581	Good	Employed	Bachelor	\$32,684	
22-06-2018	39	Mid	\$48,707	582	Good	Employed	Bachelor	\$9,741	
05-09-2018	32	Mid	\$34,817	606	Good	Employed	Bachelor	\$36,726	
21-09-2018	27	Mid	\$29,349	577	Good	Employed	Bachelor	\$31,897	
30-09-2018	53	Senior	\$45,788	635	Good	Employed	Bachelor	\$31,589	
28-10-2018	30	Mid	\$26,960	603	Good	Employed	Bachelor	\$12,517	

Data

Search

- Borrower Profile
- Credit Score Band
- Debt To Income Ratio
- Employment Status
- High Risk Customer
- Home Ownership Status
- Interest Rate
- Loan Amount
- Loan Approved
- Loan Approved Count
- Loan Duration
- Loan Purpose
- Loan Rejected Count

• Average Risk Score

1 Avg Risk Score = AVERAGE('CLEANED DATASET'[Risk Score])

Application Date	Age	Age Group	Annual Income	Credit Score	Credit Score Band	Employment Status	Education Level	Loan Amount	Loan Duration
03-01-2018	47	Senior	\$40,724	570	Good	Employed	Bachelor	\$17,627	
29-01-2018	32	Mid	\$35,498	622	Good	Employed	Bachelor	\$11,136	
09-03-2018	52	Senior	\$32,418	647	Good	Employed	Bachelor	\$32,208	
10-03-2018	44	Senior	\$26,879	649	Good	Employed	Bachelor	\$20,313	
17-03-2018	49	Senior	\$40,829	617	Good	Employed	Bachelor	\$25,478	
01-04-2018	41	Senior	\$51,707	585	Good	Employed	Bachelor	\$20,069	
08-04-2018	43	Senior	\$30,948	601	Good	Employed	Bachelor	\$25,291	
26-04-2018	43	Senior	\$49,421	609	Good	Employed	Bachelor	\$17,161	
28-04-2018	25	Mid	\$41,402	572	Good	Employed	Bachelor	\$25,900	
05-05-2018	47	Senior	\$50,345	602	Good	Employed	Bachelor	\$18,805	
01-06-2018	44	Senior	\$47,430	581	Good	Employed	Bachelor	\$32,684	
22-06-2018	39	Mid	\$48,707	582	Good	Employed	Bachelor	\$9,741	
05-09-2018	32	Mid	\$34,817	606	Good	Employed	Bachelor	\$36,726	
21-09-2018	27	Mid	\$29,349	577	Good	Employed	Bachelor	\$31,897	
30-09-2018	53	Senior	\$45,788	635	Good	Employed	Bachelor	\$31,589	
28-10-2018	30	Mid	\$26,960	603	Good	Employed	Bachelor	\$12,517	
12-11-2018	40	Senior	\$36,971	611	Good	Employed	Bachelor	\$29,198	

Data

Search

- CLEANED DATASET
- Credit Score
- Education Level
- Age
- Age Group
- Annual Income
- Application Date
- Average Loan Amount
- Avg LoanToIncome
- Avg Risk Score
- Borrower Profile
- Credit Score Band
- Debt To Income Ratio
- Employment Status

• Average of LoanToIncome

1 Avg LoanToIncome = AVERAGE('CLEANED DATASET'[Loan to Income Ratio])

Application Date	Age	Age Group	Annual Income	Credit Score	Credit Score Band	Employment Status	Education Level	Loan Amount	Loan Duration
03-01-2018	47	Senior	\$40,724	570	Good	Employed	Bachelor	\$17,627	
29-01-2018	32	Mid	\$35,498	622	Good	Employed	Bachelor	\$11,136	
09-03-2018	52	Senior	\$32,418	647	Good	Employed	Bachelor	\$32,208	
10-03-2018	44	Senior	\$26,879	649	Good	Employed	Bachelor	\$20,313	
17-03-2018	49	Senior	\$40,829	617	Good	Employed	Bachelor	\$25,478	
01-04-2018	41	Senior	\$51,707	585	Good	Employed	Bachelor	\$20,069	
08-04-2018	43	Senior	\$30,948	601	Good	Employed	Bachelor	\$25,291	
26-04-2018	43	Senior	\$49,421	609	Good	Employed	Bachelor	\$17,161	
28-04-2018	25	Mid	\$41,402	572	Good	Employed	Bachelor	\$25,900	
05-05-2018	47	Senior	\$50,345	602	Good	Employed	Bachelor	\$18,805	
01-06-2018	44	Senior	\$47,430	581	Good	Employed	Bachelor	\$32,684	
22-06-2018	39	Mid	\$48,707	582	Good	Employed	Bachelor	\$9,741	
05-09-2018	32	Mid	\$34,817	606	Good	Employed	Bachelor	\$36,726	
21-09-2018	27	Mid	\$29,349	577	Good	Employed	Bachelor	\$31,897	
30-09-2018	53	Senior	\$45,788	635	Good	Employed	Bachelor	\$31,589	
28-10-2018	30	Mid	\$26,960	603	Good	Employed	Bachelor	\$12,517	
12-11-2018	40	Senior	\$36,971	611	Good	Employed	Bachelor	\$29,198	

Data

Search

- CLEANED DATASET
- Credit Score
- Education Level
- Age
- Age Group
- Annual Income
- Application Date
- Average Loan Amount
- Avg LoanToIncome
- Avg Risk Score
- Borrower Profile
- Credit Score Band
- Debt To Income Ratio
- Employment Status

DAX Calculated Column

• Loan Status

1 Loan Status = If('CLEANED DATASET'[Loan Approved]=1,"Approved","Rejected")

ings Account Balance	Monthly Income	Monthly Income Category	Net Worth	Interest Rate	Loan Approved	Risk Score	Risk Score Label	Loan Status
\$886	\$3,393.66667	Low Income	\$5,205	21%	0	52	Medium Risk	Rejected
\$1,346	\$2,958.16667	Low Income	\$2,705	16%	0	48	Medium Risk	Rejected
\$2,926	\$2,701.5	Low Income	\$3,13,319	15%	0	50	Medium Risk	Rejected
\$4,029	\$2,239.91667	Low Income	\$71,090	20%	0	50	Medium Risk	Rejected
\$2,267	\$3,402.41667	Low Income	\$34,774	18%	0	49	Medium Risk	Rejected
\$3,815	\$4,308.91667	Low Income	\$2,253	27%	0	52	Medium Risk	Rejected
\$7,167	\$2,579	Low Income	\$78,660	22%	0	49	Medium Risk	Rejected
\$10,050	\$4,118.41667	Low Income	\$9,540	26%	0	51	Medium Risk	Rejected
\$9,920	\$3,450.16667	Low Income	\$4,623	24%	0	50	Medium Risk	Rejected
\$13,605	\$4,195.41667	Low Income	\$47,177	23%	0	52	Medium Risk	Rejected
\$673	\$3,952.5	Low Income	\$25,415	23%	0	50	Medium Risk	Rejected
\$2,817	\$4,058.91667	Low Income	\$69,075	24%	0	50	Medium Risk	Rejected
\$3,099	\$2,901.41667	Low Income	\$2,880	26%	0	54	Medium Risk	Rejected
\$11,187	\$2,445.75	Low Income	\$2,007	34%	0	54	Medium Risk	Rejected
\$12,430	\$3,815.66667	Low Income	\$23,478	19%	0	52	Medium Risk	Rejected
\$14,928	\$2,246.66667	Low Income	\$48,358	20%	0	52	Medium Risk	Rejected
\$2,250	\$3,080.91667	Low Income	\$3,965	23%	0	47	Medium Risk	Rejected
\$8,040	\$2,409	Low Income	\$2,610	18%	0	51	Medium Risk	Rejected

Data

Search

- Borrower Profile
- Credit Score Band
- Debt To Income Ratio
- Employment Status
- High Risk Customer
- Home Ownership Status
- Interest Rate
- Loan Amount
- Loan Approved
- Loan Approved Count
- Loan Duration
- Loan Purpose
- Loan Rejected Count
- Loan Status
- Loan to Income Ratio

8. Analysis and Visualizations (Power BI)

Dashboard Visuals

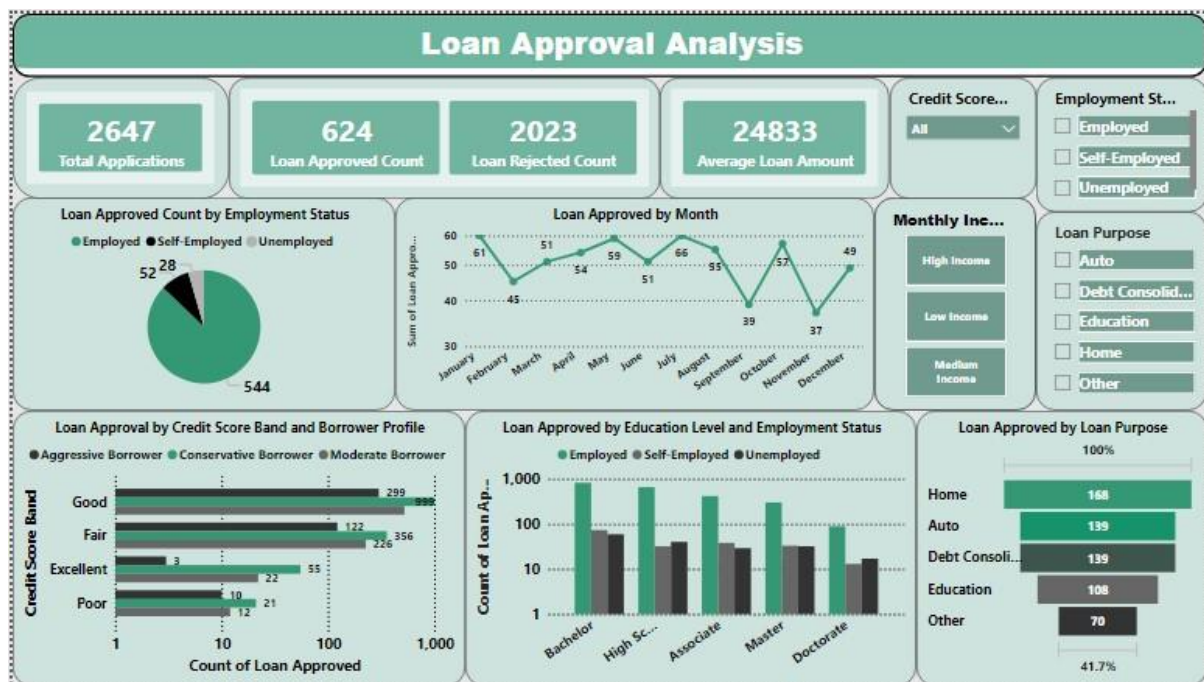
- KPI Cards
- Bar / Column Charts
- Pie Charts
- Line Charts
- Donut Charts
- Funnel Charts

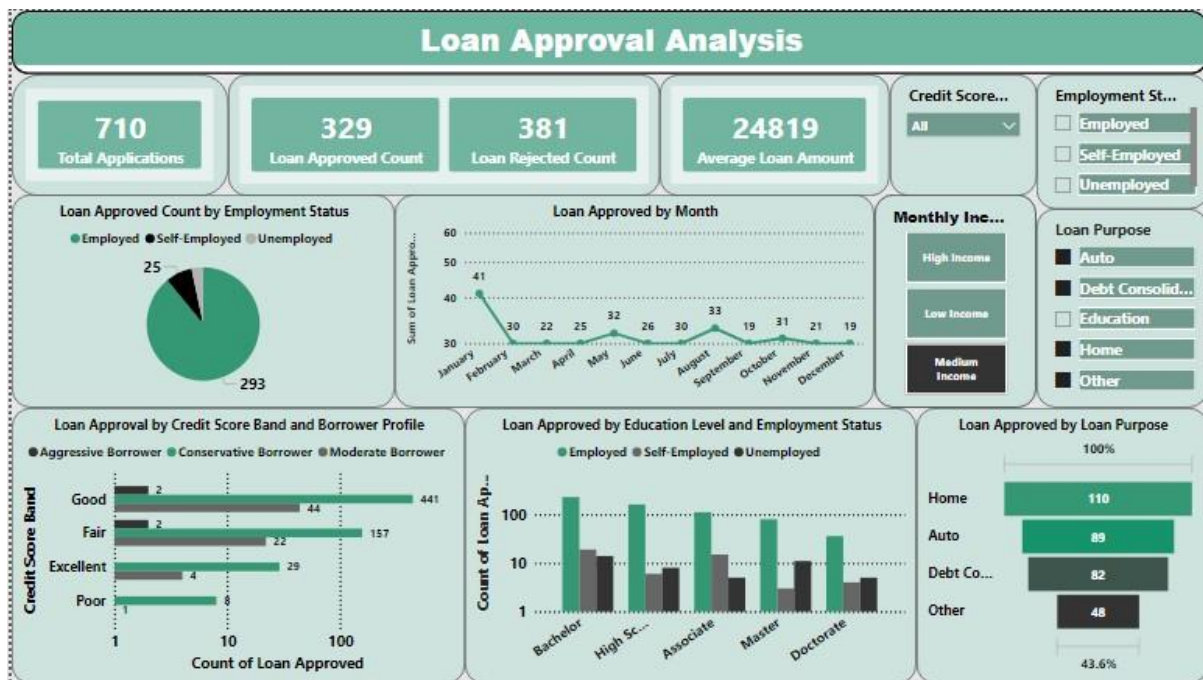
Slicers Used

- Employment Status
- Loan Purpose
- Monthly Income
- Risk Category

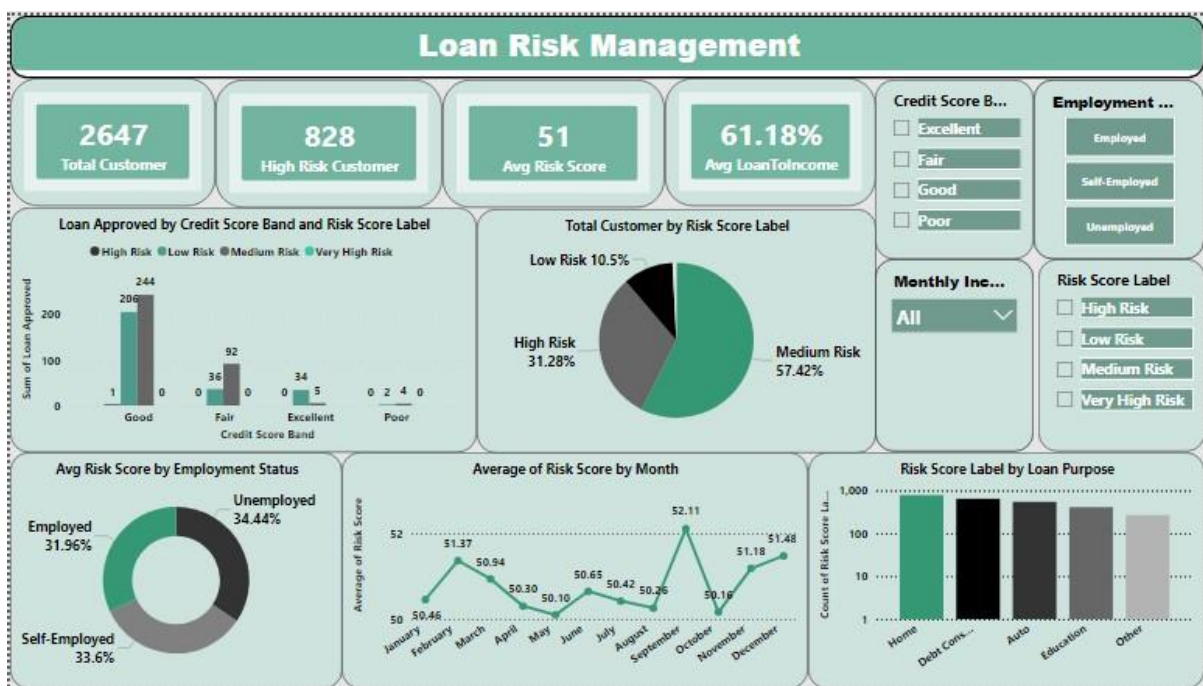
SCREENSHOT OF DASHBOARD /REPORT

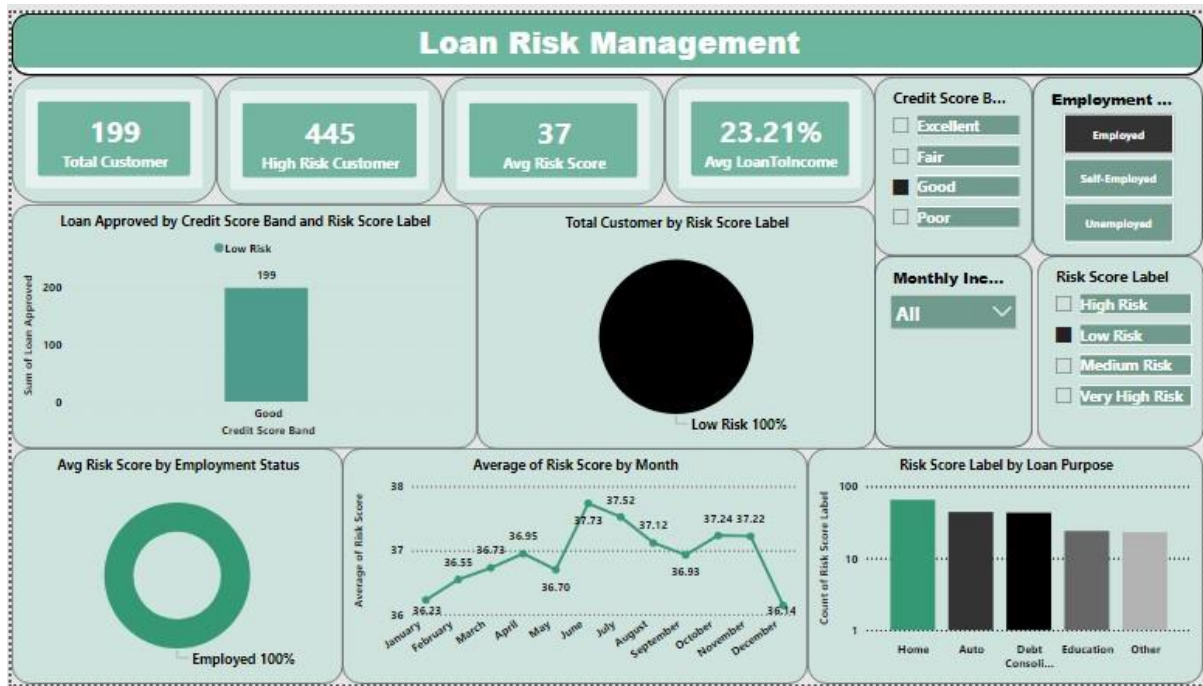
Dashboard 1: Loan Approval Analysis





Dashboard 2: Loan Risk Management





9. Insights & Conclusions

Key Findings

Descriptive Analysis

- Total **2,647 loan applications** were analysed.
- Loan approvals are lower compared to rejections (**624 approved, 2,023 rejected**).
- Average loan amount is around **\$24,833**.
- Most approved loans are from **employed customers**.
- Customers with **good credit score, low risk score, and low debt-to-income ratio** have higher chances of approval.

Diagnostic Analysis

- Loan rejection is significantly higher compared to loan approval.
- The main reason for rejection is **low credit score and high-risk score** among customers.
- Customers with poorly maintained **savings accounts** also show higher rejection rates.
- An imbalance between **annual income, debt-to-income ratio, and loan-to-income ratio** increases the chances of loan rejection.
- Overall, customers with weak financial stability are more likely to face loan rejection.

Predictive Analysis

- Out of **2,647 customers**, around **828 customers** are identified as **high-risk customers**.
- Due to the high number of **high-risk customers**, **loan rejection** may continue to increase in the future.
- Customers with **low credit score**, **high debt-to-income ratio**, **unstable monthly income**, and **poor financial balance** are more likely to face **loan rejection**.
- The bank can use these factors to predict **future loan approval chances** and identify **high-risk borrowers** earlier.

Prescriptive Analysis

- The bank should focus on customers with better credit scores and stable income.
- High-risk customers can be given financial guidance before loan approval.
- Customers should maintain a balanced debt-to-income ratio and savings account.
- The bank can improve approval accuracy by monitoring risk score and financial behaviour.
- Loan eligibility rules can be optimized to reduce future loan defaults and rejection rates.

10. Conclusions

The integration of Excel and Power BI proved highly effective for performing end-to-end banking data analysis. The project successfully transformed raw customer loan data into meaningful business insights through data cleaning, modelling, DAX calculations, and dashboard visualizations.

The interactive dashboard helps users:

- Understand customer financial behaviour
- Analyse loan approval patterns
- Identify risky customers
- Improve banking decision-making
- Support risk management strategies

This project enhanced practical knowledge in:

- Excel Data Cleaning
- Power Query Transformation
- Power BI Data Modelling
- DAX Measures
- Interactive Dashboard Design
- Business Intelligence and Analytics